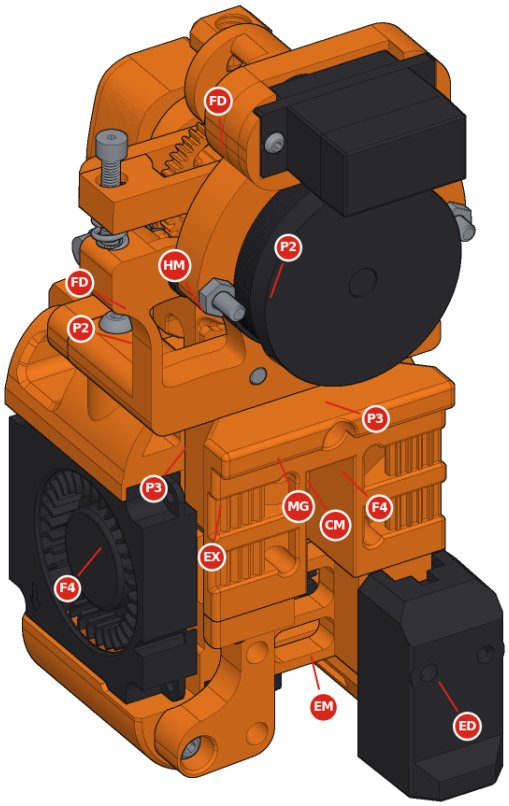
Electronics manual (all seven frames, 1–8 tools)

Boards, power, wiring, Klipper and MedusaHC · all seven frames

MedusaHC toolhead · Complete toolhead with a tool

The toolhead with the feeder on top and a hotend tool held by the magnets and guide pins.



Real parts: Sergei (Irbis3D) MedusaHC V0.2 STEP files (GPL-3.0), real size.
cad/designs/duender-large/render_toolhead.py

You need

- EX 1x Extruder Main
- CM 1x MGN9H Mount Main (frame ch. 7)
- EM 1x MGN9H Mount EDDY ADXL
- ED 1x BTT Eddy probe
- F4 2x 4010 radial fan
- MG 1x 12×3 mm magnet
- P3 2x M3×30 guide pin
- P2 2x M3×20 guide pin
- FD 1x feeder (Chapter 5)
- HM 1x hotend tool, picked up

October 2026

Board facts from BIGTREETECH's manuals and configs, Klipper's board configs and Sergei's MedusaHC V0.2 (GPL-3.0).

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Electronics manual (all seven frames, 1–8 tools)

For every frame in this project: C300 cut, C300 from CR-10, C300 from CR-10 V2, C400 cut, C400 from CR-10 S4, C500 cut, C500 from CR-10 S5, with **MedusaHC** rear docks and any number of tools the frame takes (toolhead manual: up to 5 on the C300s, 6 on the C400s, 8 on the C500s).

Before you start: the frame manual and the toolhead manual are finished, and everything moves freely by hand.

What this manual covers: choosing the boards for your tool count, the power supplies (and the AC bed on the C400 and C500), wiring, Klipper firmware, the configuration (Sergei's MedusaHC files, changed for these frames and for rear docks), and the first start and calibration.

> ⚠️ **Mains safety.** This manual connects mains-powered supplies and, on the C400 and C500, a mains bed heater. Earth everything metal and keep mains in a closed box. **If you are not sure, have a qualified electrician check it before you switch on.**

Chapters

Ch	Chapter
00	Choosing the boards
01	Power
02	Wiring
03	Klipper firmware
04	Configuration
05	First start and calibration

The short answer

Tools	Boards	Main 24 V supply
1–4	**BTT Manta M8P V2.0 + CB2** (Sergei's set-up), or a BTT Octopus V1.1 + Raspberry Pi	350 W
5–6	the same + **1 × BTT SKR Mini E3 V3**	500 W *(est.)*
7–8 (C500 only)	the same + **2 × BTT SKR Mini E3 V3**	600 W *(est.)*

The bed has its own supply (or a relay for an AC bed): Chapter 1. The **donor CR-10 boards** can work
as extra boards, but the CR-10, S4 and S5 boards are 12 V and the tools' heaters are 24 V: see Chapter 0.

Sources: BTT's manuals and configs for the M8P V2.0, Octopus and SKR Mini E3 V3, Klipper's board configs, and Sergei's (Irbis3D) MedusaHC V0.2 README and configuration (GPL-3.0). Every number is listed with its source in research/ELECTRONICS-FACTS.md.

Chapter 0: Choosing the boards

Contents · Next: Power →

This manual covers three BTT boards: **Manta M8P V2.0**, **Octopus V1.1 / Pro** and **SKR Mini E3 V3.0**. Any mix of them works, because Klipper runs several boards as one printer. BTT's pin diagrams, manuals and schematics for each board are linked in Chapter 2.

What needs a port

Group	Needs
Frame	6 stepper drivers: A, B, Z × 3 (independent, for z_tilt), extruder (NEMA14); bed heater + bed thermistor
Toolhead	extruder driver (above), part-cooling fans (2 × 4010 on one output), toolhead/feeder switch, Eddy probe (CAN or USB), servo (MG90S, PWM pin, 5 V)
Per tool	heater output (MOSFET), thermistor input, fan output (PWM or a constant 24 V port), dock switch input (the switch is on the dock)

The tools' heaters are Bambu-style ceramic heaters, **24 V 48 W** (2 A each). X and Y home sensorlessly (the drivers' DIAG lines use two stop inputs); Z homes on the Eddy.

The boards

	Manta M8P V2.0	Octopus V1.1 / Pro	SKR Mini E3 V3.0
Drivers	8	8 (one slot has two parallel outputs)	4 (Z has two parallel outputs)
Heaters	HB + HE0–HE3; HE 5.5 A, HB 10 A	bed + HE0–HE3	HB + E0
Thermistors	TB + TH0–TH3	TB + T0–T3	THB + TH0
Fans	FAN0–FAN6 (5/12/24 V selectable) + Pi fan + 1 always-on	6 PWM + 2 always-on	FAN0–FAN2
Stop inputs	6 + the PF10 header	8 + probe port	X, Y, Z, E0 + probe
Servo	dedicated Servo port	BLTouch header (no servo port)	BLTouch/probe header
Power	VIN 12/24 V, BED IN 12/24 V; **heaters + drivers + fans < 12 A**	24 V (**logic needs > 14.1 V**); separate motor and bed inputs	12/24 V
Host	CB1/CB2/CM4 on the board	separate (Raspberry Pi)	separate (USB to the same host)

How many tools each set runs

Tools	Board set	Why
1–4	**M8P V2.0** or **Octopus** alone	both have 4 hotend heaters and 4 tool thermistors; the 4 heaters run out first. On the M8P the stop inputs run out too: 2 are the X/Y DIAG lines, 4 are dock switches
5–6	M8P or Octopus + **1 × SKR Mini E3 V3**	the SKR Mini adds 2 heaters (E0 + HB), 2 thermistors, 3 fans and its stop inputs: **2 tools**
7–8	M8P or Octopus + **2 × SKR Mini E3 V3**	2 tools each. 8 docks only fit on the C500s (toolhead manual)

The SKR Mini E3 V3 isn't a main board here: its 4 drivers can't run three independent Z motors.

Donor boards

Donor	Board	Supply	Use here
CR-10 (2017)	Melzi-type, ATmega1284P	12 V	12 V rule below
CR-10S / S4 / S5	Creality ATmega2560 board	12 V	12 V rule below; **check the board label**: later units changed boards
CR-10 V2	Creality "silent" board, TMC2208	24 V	hotend + bed outputs = 2 tools, **once you have confirmed its pins** from the board's label and its Marlin pins file. Klipper has no stock config for it

12 V rule: the tools' heaters are 24 V. A 12 V board's heater output gives a 24 V heater a quarter of its power. Don't run a 12 V donor board at 24 V unless its own documentation says it can. A 12 V donor board can drive 24 V heaters through an external MOSFET module (Chapter 1, Step 1.4). *(Design rule from the board specs, not a tested build.)*

Other builders have done this: one runs 6 heads on an Octopus + 2 Creality boards, another got 7 hotends from 3 old boards.

Our pick

Up to 4 tools: the **M8P V2.0 + CB2**. That's Sergei's set-up, so his configuration files fit with the fewest changes. The **Octopus** works the same way with a Raspberry Pi. **More tools:** add one **SKR Mini E3 V3** for every 2 extra tools. It runs on 24 V, so it needs no MOSFET modules.

Chapter 1: Power

← Choosing the boards · Contents · Next: Wiring →

> ⚠️ **Mains safety.** The supplies, and the AC bed on the C400 and C500, run on mains voltage. **Earth everything metal**: the supplies, the frame and the bed plate. Put every mains connection in a closed box with a fused, switched inlet. **If you are not sure, have a qualified electrician check it before you switch on.**

How the power is split

Supply	Feeds	Voltage
Main supply	the board(s): logic, stepper drivers, fans, servo, Eddy, **the tool heaters**	**24 V**: the tools' Bambu-style ceramic heaters are 24 V 48 W
Bed	the bed heater only	the bed's own voltage: 12 V or 24 V DC from a supply, or **mains (AC)** through a solid-state relay

Join the 0 V (−) of every DC supply with a short, thick wire, so all the boards share one ground. **Never connect two supplies' + outputs together** (a MedusaHC user found paralleled switching supplies fight each other).

Step 1.1: The main 24 V supply

Sergei: one 350 W supply "is reliably sufficient for 2 hotends. For 3 hotends, it can be enough with a high-quality PSU. For 4 hotends, it is definitely not enough". His 4-tool printer has a second 350 W supply for the bed, so the main one feeds only the board, motors, fans and tools.

Tools	Main 24 V supply ***(est.: (100 W + 48 W per tool) × 1.25)***
1–4	**350 W** (Sergei's, with the bed on its own supply)
5–6	**500 W**, or a second 24 V supply for the extra board (0 V joined)
7–8	**600 W**, or one 24 V supply per board set (0 V joined)

Do:

1. Use a branded supply (Mean Well or similar), **24 V**. The CR-10 V2's 24 V 350 W supply works for up to 4 tools.
2. Set its output to **24.0 V** with its trimmer, measured with a meter.
3. Wire mains to it through the fused, switched inlet. **Earth to its earth terminal.**
4. Run **24 V** to each board's power input, with wire rated for the current (1.5 mm² / 16 AWG for the main board).

⚠️ **CR-10, S4 and S5 donors:** their supplies are **12 V**. Keep one for a 12 V bed (Step 1.2), but don't use it for the main supply: 24 V heaters run at a quarter power on 12 V, and the Octopus's logic needs more than 14.1 V.

⚠️ **M8P V2.0 limit:** BTT rates the heater cartridges + drivers + fans at **less than 12 A in total**. 4 tools heating at once draw 4 × 2 A = 8 A. Don't add more heaters to the M8P: tools 5+ go on their own board (Chapter 0).

Step 1.2: The bed

Frame	Bed (frame manual)	How to power it
C300 cut, from CR-10	CR-10 bed, **12 V**	the CR-10's 12 V supply → the board's **bed input** (M8P BED IN or Octopus bed input: both take 12 V or 24 V, separate from the logic)
C300 from CR-10 V2	CR-10 V2 bed, **24 V**	a 24 V supply for the bed only (the V2's own one, if the main supply is new)
C400 cut, from CR-10 S4	**AC 400 bed** (mains)	solid-state relay, Step 1.3
C500 cut, from CR-10 S5	510 plate, 12 V heater removed: **AC heater pad** (mains)	solid-state relay, Step 1.3

Do (DC bed):

1. Wire the bed supply's + and − to the board's **bed input**, then the bed heater to **HB**. Use thick wire (2.5 mm² / 14 AWG): a 12 V bed draws a lot of current.
2. Join the bed supply's 0 V to the main supply's 0 V.
3. Earth the bed plate (a ring terminal under one of its screws, to the supply's earth).

✅ **Check:** with a meter, the bed input reads the bed supply's voltage, the board input reads 24 V, and the two 0 V terminals read 0 V between them.

Step 1.3: An AC bed through a solid-state relay (C400, C500)

The board switches the relay, and the relay switches the mains to the bed heater. The board never touches mains.

You need	Qty
Solid-state relay: DC control (3–32 V), AC load, zero-cross, rated well above the bed current, on a heatsink	1
Bed heater pad or AC bed with a **thermal fuse** (thermal cut-off) bonded to it	1
Earth wire to the bed plate	1
Fuse for the bed circuit, sized for the bed	1

Do:

1. Mount the relay on its heatsink, inside the closed mains box.
2. **Control side:** the board's **HB** output → the relay's DC input (+ to +). Klipper switches it like a DC bed.
3. **Load side:** mains live → bed fuse → relay → bed heater → neutral. Thermal fuse in series with the heater, on the bed.
4. **Earth the bed plate** to the mains earth.

5. In Klipper (Chapter 4), set the bed heater's max_power for your heater and keep [verify_heater heater_bed] on. A failed-on relay is the main AC-bed risk: the thermal fuse is the backstop.

⚠️ Have a qualified electrician check the mains side **before the first switch-on**.

Step 1.4: A 12 V donor board on 24 V heaters (only if you use one)

Chapter 0 explains why a 12 V donor board can't drive the tools' 24 V heaters directly. If you use one anyway as an extra board, switch each 24 V heater through an **external MOSFET module** (heater-rated, with a trigger input):

1. The donor board's heater output (12 V) → the module's **trigger** input.
2. The main 24 V supply → the module's **power** input. The module's output → the tool heater.
3. Join the donor board's 0 V to the main 0 V.

(This is a design rule from the board specs, not a tested build. The SKR Mini E3 V3 (24 V) avoids it.)

Step 1.5: Before anything is plugged into a board

✅ **Check:**

- Every supply's output voltage is right, measured with a meter, **before** it is connected to a board.
- All 0 V terminals are joined; no two + outputs are joined.
- Earth reaches every supply, the frame and the bed plate. Check it with the meter's continuity setting.
- The mains box is closed and nothing bare is showing.

Chapter 2: Wiring

← Power · Contents · Next: Klipper firmware →

BTT's own documents for each board are on BIGTREETECH's GitHub: pin diagram (where every port is on the board), user manual (jumpers, wiring examples), schematic. **Open the pin diagram for your board before you start**, and keep it next to the printer:

Board	BTT repository	Pin diagram	User manual	Schematic
Manta M8P V2.0	bigtreetech/Manta-M8P	in the user manual	V2.0 User Manual	V2.0 SCH
Octopus V1.0 / V1.1	bigtreetech/BIGTREETECH-OCTOPUS-V1.0 (covers V1.1)	in the user manual	Octopus EN manual	Octopus SCH
Octopus Pro	bigtreetech/BIGTREETECH-OCTOPUS-Pro	Octopus Pro PIN	in the repository	Pro V1.1 SCH
SKR Mini E3 V3.0	bigtreetech/BIGTREETECH-SKR-mini-E3	V3.0 PIN	V3.0 user manual	V3.0 SCH

The pins in the tables below come from BTT's and Klipper's configuration files for these boards (generic-bigtreetech-manta-m8p-V2_0.cfg in BTT's M8P repository; Klipper's generic-bigtreetech-octopus-v1.1.cfg and generic-bigtreetech-skr-mini-e3-v3.0.cfg). Find each port on the pin diagram by its **name** (Motor1, HE0, FAN2…). If your board is a different revision (SKR Mini E3 **V3.0.1**, M8P **V1.x**), use **its** pin diagram and config from the same repositories: some pins differ.

Power off and unplugged for this whole chapter. Wire **one cable at a time** and tick it off in the table for your board. The ports are the ones **Chapter 4's configuration** uses: if you change one here, change it there too.

Step 2.1: Mount the boards

Do:

1. Mount the main board, and any extra board, on stand-offs in a ventilated box under or behind the printer. Keep the **mains box separate** (Chapter 1).
2. Point a fan across the stepper drivers.
3. Set the jumpers **before** fitting the drivers (main board):
    - **UART** mode for every driver slot in use (TMC2209).
    - **Sensorless X/Y homing:** fit the **DIAG** jumper on the **A and B** slots only. **Leave DIAG off** on every other slot: their stop inputs carry the dock switches.
    - Driver supply (VMOT) **24 V**, and fan voltage jumpers at **24 V** (all the fans here are 24 V).

⚠️ Andizzle's Gridfinity base case has a bin for an **SKR Mini E3 V3**, not for the M8P or the Octopus. An SKR Mini used as an extra board fits it.

Step 2.2: Main board: Manta M8P V2.0 (Sergei's ports)

☐	Function	Port (pin)	Klipper
☐	A motor (right, belt A)	Motor1	[stepper_x]
☐	B motor (left, belt B)	Motor2	[stepper_y]
☐	Z front, Z left, Z right	Motor3, Motor4, Motor6	[stepper_z], [stepper_z1], [stepper_z2]
☐	Extruder (NEMA14 on the toolhead)	Motor5	[extruder] (shared by every tool)
☐	Bed heater / bed thermistor	HB / TB	[heater_bed]
☐	Tool 0–3 heaters	HE0, HE1, HE2, HE3	[extruder] … [extruder3]
☐	Tool 0–3 thermistors	TH0, TH1, TH2, TH3	the same sections
☐	Tool 0–3 fans (3010)	FAN0, FAN1, FAN2, FAN3 (PF7, PF9, PF6, PF8)	[heater_fan fan1] … fan4
☐	Part-cooling fans (2 × 4010, together)	FAN4 (PA4)	[fan]
☐	Board/case fan	FAN5 or FAN6	[controller_fan]
☐	Dock 0–3 switches	Motor3–Motor6 stop inputs (PF2, PF1, PF0, PC15)	pin_t0 … pin_t3
☐	Toolhead (feeder) switch	PF10 header	pin_e
☐	Servo (MG90S)	Servo port (PE9)	[servo my_servo]
☐	Eddy probe	CAN (or USB to the CB2)	[probe_eddy_current]

Step 2.2 (alternative): Main board: Octopus V1.1 / Pro

☐	Function	Port (pin)	Klipper
☐	A, B motors	Driver0, Driver1 (DIAG jumpers on)	[stepper_x], [stepper_y]
☐	Z front, Z left, Z right	Driver2 (**MOTOR2_1 only**: 2_2 is the same driver), Driver3, Driver5	[stepper_z], z1, z2
☐	Extruder	Driver4	[extruder]
☐	Bed heater / thermistor	bed output (PA1) / TB (PF3)	[heater_bed]
☐	Tool 0–3 heaters	HE0–HE3 (PA2, PA3, PB10, PB11)	[extruder] … [extruder3]
☐	Tool 0–3 thermistors	T0–T3 (PF4, PF5, PF6, PF7)	the same sections
☐	Tool 0–3 fans	FAN1–FAN4 (PE5, PD12, PD13, PD14)	[heater_fan] × 4
☐	Part-cooling fans	FAN0 (PA8)	[fan]
☐	Board/case fan	FAN5 (PD15)	[controller_fan]
☐	Dock 0–3 switches	stop inputs PG10, PG11, PG12, PG13	pin_t0 … pin_t3

☐	Function	Port (pin)	Klipper
☐	Toolhead (feeder) switch	stop input PG14	pin_e
☐	Servo	BLTouch header, control pin (PB6)	[servo my_servo]
☐	Eddy probe	CAN (RJ11) or USB to the host	[probe_eddy_current]
☐	Host	Raspberry Pi (USB)	[mcu]

⚠️ The Octopus needs **more than 14.1 V** for its logic: run it on 24 V.

⚠️ The Octopus has no servo port of its own: power the MG90S from a **24 V→5 V converter** if the header's 5 V sags under the servo's load (MedusaHC README).

Step 2.3: Extra board for tools 4–5 (and 6–7): SKR Mini E3 V3

One SKR Mini E3 V3 runs **two more tools**. Use a second one for tools 6–7 (C500 only).

☐	Function	Port (pin)	Klipper (board named sk1, second one sk2)
☐	Power	24 V from the main supply (or its own, 0 V joined)	—
☐	Tool 4 heater / thermistor	E0 (PC8) / TH0 (PA0)	[extruder4]
☐	Tool 5 heater / thermistor	**HB** (PC9) / THB (PC4): the bed output drives a 2 A tool heater easily	[extruder5]
☐	Tool 4, 5 fans	FAN0 (PC6), FAN1 (PC7)	[heater_fan fan5], fan6
☐	Dock 4, 5 switches	X-STOP (PC0), Y-STOP (PC1)	pin_t4, pin_t5
☐	Link	USB to the host	[mcu sk1]

Leave its stepper drivers unused.

Step 2.4: The cables to the tools and the toolhead

Every tool has its **own** cable: heater (2 wires), thermistor (2) and fan (2). The cable moves with the tool while the toolhead carries it.

Do:

1. Run each tool's cable **up from the tool to the top of the frame**, then along to the boards. Fix it to the frame with Sergei's wire holders (Wire_Holder M6-M10, Wire Holder M6-M6, Wire_Holder PU4: pick the one for your fittings) or a cable chain.
2. Give each cable a loop long enough for the tool to reach **every corner of the bed** and its dock. Check it by hand with the power off.
3. Do the same for the **toolhead**: extruder motor (4), part fans (2), feeder switch (2), servo (3), Eddy (CAN 4, or USB).
4. Strain-relieve every cable at both ends. **Label every cable end** (T0 heater, T0 thermistor, …).
5. The **dock switches** are fixed: run their cables along the back brace to the boards.

✅ **Check before Chapter 3:**

- Every row in your board's table is ticked.
- No bare copper at any terminal; tug every wire in its terminal.
- The motor cables go to the slots in the table.
- **Supply polarity checked with the meter**, at the board end of each power cable.
- The M8P's VUSB jumper is off (unless you are flashing from USB power).

Chapter 3: Klipper firmware

← Wiring · Contents · Next: Configuration →

The **host** (a computer: the M8P's CB1/CB2/CM4, or a Raspberry Pi for the Octopus) runs Klipper, Moonraker and Mainsail. **Every board** (MCU) runs Klipper firmware built for it. All the firmware settings below are from the board makers' documents (BTT's M8P V2.0 manual and configs, Klipper's board configs, Voron's guides).

Step 3.1: First power-on, nothing moving

Do:

1. **Unplug the motors** at the board for this first test (or leave the drivers out).
2. Switch on. Watch and smell: **switch off at once** if anything smokes or smells hot.
3. Measure: **24 V** at each board's power input, **5 V** at the host.

✅ **Check:** every board's power LED is on, and nothing is warm except the supplies.

Step 3.2: The host

Do (M8P with a CB1/CB2):

1. Write BTT's CB1/CB2 image to a microSD card (Raspberry Pi Imager, "Use custom"). For Wi-Fi, edit system.cfg on the card's FAT32 partition (WIFI-SSID and PASSWORD).
2. Boot it, find its IP address on your router (hostname BTT-CB1, Fluidd or Mainsail), and log in over SSH. BTT's defaults: CB1 biqu / biqu; CM4 pi / raspberry. **Change the password.**

Do (Octopus): install Raspberry Pi OS Lite on the Pi, then Klipper, Moonraker and Mainsail with **KIAUH**.

✅ **Check:** Mainsail opens in your browser at the host's IP address.

Step 3.3: Build and flash the main board

Do:

1. On the host: cd ~/klipper && make menuconfig, with **Enable extra low-level configuration options** ticked:

Board	Processor	Bootloader offset	Clock	Communication
Manta M8P V2.0	STM32H723	**128KiB bootloader**	**25 MHz crystal**	USB (on PA11/PA12), or CAN bus (on PD0/PD1)
Octopus V1.1 / Pro (F446)	STM32F446	**32KiB bootloader**	**12 MHz crystal**	USB (on PA11/PA12)
SKR Mini E3 V3.0 (extra board)	STM32G0B1	**8KiB bootloader**	**8 MHz crystal**	USB (on PA11/PA12)

1. Save, then make clean && make. The firmware is ~/klipper/out/klipper.bin.

2. Copy it to a microSD card as **firmware.bin**, put the card in the board, and press reset or power on. The board renames the file to FIRMWARE.CUR when it has flashed.
3. ls /dev/serial/by-id/ lists the board as a Klipper device. Copy that path.

⚠️ **One build at a time:** make menuconfig keeps one setting. Build, copy and flash each board before you change the settings for the next.

⚠️ **SKR Mini E3 V3:** make flash doesn't work on it: use the SD card. Some boards ship with an **STM32G0B0** instead of the G0B1: it works the same over USB, but has no CAN. Read the chip.

⚠️ **CAN (optional):** Sergei runs his M8P and the Eddy on CAN. CAN needs the board's CAN termination jumper and a CAN interface on the host. Start on **USB**: it's simpler, and changing later only changes the [mcu] line.

Step 3.4: Extra boards

Do:

1. Build and flash each SKR Mini E3 V3 as in Step 3.3 (8KiB bootloader, 8 MHz, USB).
2. Plug it into the host by USB. ls /dev/serial/by-id/ now lists two (or three) Klipper devices: **write down which is which**. Unplug one and list again if you are not sure.
3. **Donor 8-bit board** (if you use one, Chapter 0): build for its AVR chip (CR-10 2017: **atmega1284p**; CR-10S type: **atmega2560**), and flash it as Klipper's config for that printer describes (printer-creality-cr10-2017.cfg, printer-creality-cr10s-2017.cfg). Its heaters and drivers only work with the **printer's supply on**, not from USB power alone.

Step 3.5: The Eddy probe

Do: follow BTT's Eddy instructions for its firmware, over **USB** (simplest) or **CAN** (Sergei's). Write down its serial path or CAN UUID for Chapter 4.

✅ **Check (end of chapter):** Mainsail shows every board (MCU) connected, with no "unable to connect" errors, once Chapter 4's [mcu] sections are in.

Chapter 4: Configuration

← Klipper firmware · Contents · Next: First start and calibration →

Start from **Sergei's working files**, not from a blank page. They are for his 4-tool Duender on a Manta M8P V2.0 with front docks. This chapter lists what changes for **these frames**, for **rear docks (V-Back)**, for the **Octopus**, and for **tools 4–7**.

Step 4.1: Install MedusaHC and the example files

Do:

1. On the host, install the MedusaHC core (MedusaHC README):

```
bash -c "$(curl -fsSL https://raw.githubusercontent.com/Irbis3D/MedusaHC/main/install-online.sh)"
```

1. It copies the controller and pin_watch.py, and makes ~/printer_data/config/MedusaHC/ with MHC_config.cfg, MHC_variables.cfg and MHC_macros.cfg. It doesn't touch printer.cfg.
2. Download Sergei's printer.cfg, sensorless.cfg and eddy_config.cfg (the same folder) as your starting printer.cfg and includes.
3. MHC_config.cfg has the complete [extruder] section. **Delete any other [extruder]** from printer.cfg. The [include MedusaHC/MHC_config.cfg] line is already in Sergei's printer.cfg.

Step 4.2: The boards

```
[mcu]
serial: /dev/serial/by-id/usb-Klipper_stm32h723xx_...   # your main board, from Step 3.3 (or
canbus_uuid: ...)

[mcu sk1]                                               # only with an extra SKR Mini E3 V3 (tools
4-5)
serial: /dev/serial/by-id/usb-Klipper_stm32g0b1xx_...

[mcu sk2]                                               # a second one (tools 6-7, C500)
serial: /dev/serial/by-id/usb-Klipper_stm32g0b1xx_...
```

Octopus instead of the M8P: change every pin in printer.cfg and MHC_config.cfg to the Octopus ports of Chapter 2 (Step 2.2, alternative). Use Klipper's generic-bigtreetech-octopus-v1.1.cfg for the driver pins.

Step 4.3: The frame

Section	Setting	Value
[stepper_x], [stepper_y]	rotation_distance	**32** (16T pulleys, Frame manual Step 6.1: teeth × 2)
	position_max	**measure**: jog to the far ends slowly (Chapter 5)

Section	Setting	Value
[stepper_z], [stepper_z1], [stepper_z2]	three sections	Sergei's printer has two Z motors. Add [stepper_z2] on Motor6 (M8P: step_pin: PG9, dir_pin: PD7, enable_pin: !PG11, [tmc2209 stepper_z2] uart_pin: PG10)
	rotation_distance	**8** (T8 × 8 lead screws)
[z_tilt]	z_positions	your lead-screw centres from **Frame manual Step 5.10**, in the stepper order of that table
	points	three points inside the bed, near the screws
[bed_mesh]	mesh_min, mesh_max	inside the bed: C300 **300**, C400 **400**, C500 **500** mm square, less the probe offset
[heater_bed]	max_power	1.0 for a DC bed; for an AC bed through a relay, set it for your heater (Chapter 1)

Step 4.4: Rear docks (V-Back)

File	Setting	Value
MHC_variables.cfg	variable_tools_direction	**-1** (rear docks)
	variable_x_t0 … variable_x_t3	each dock's X, **measured** (Chapter 5). Pitch **58 mm** (toolhead manual, Chapter 3)
	variable_y_safe, variable_y_latch	**measured** near the back end of Y (Chapter 5); Sergei's are front-dock values
	variable_y_prime, variable_y_brush, variable_x_prime_shift	measured at your bins
	variable_servo_open_angle, variable_servo_close_angle	set for the **lever feeder**: it turns the servo the **other way** from the old cam (toolhead manual, Step 5.2). Find both angles slowly in Chapter 5
MHC_variables.cfg (_GLOBAL_STATE)	variable_max_tool	your number of tools
sensorless.cfg and the START_PRINT, END_PRINT, PAUSE, RESUME, CANCEL_PRINT macros	park positions	**toward the front**, away from the docks (Sergei's park at the back)

Step 4.5: Tools 4–7 on the extra boards

Copy the T3 blocks in MHC_config.cfg once per extra tool, with the extra board's pins (Chapter 2, Step 2.3). Tool 4 on sk1:

```
[extruder4]
step_pin: PG13              # the same shared extruder motor as every tool (main board, Motor5)
dir_pin: PG12
enable_pin: !PG15
```

```
# ...the rest as [extruder3]...
heater_pin: sk1:PC8         # SKR Mini E3 V3 E0
sensor_pin: sk1:PA0         # TH0
sensor_type: ATC Semitec 104GT-2

[heater_fan fan5]
pin: sk1:PC6                # FAN0
heater: extruder4
heater_temp: 50.0

[verify_heater extruder4]
# as [verify_heater extruder3]

[gcode_macro T4]
# as [gcode_macro T3], for tool 4
```

- **Tool 5** on sk1: heater_pin: sk1:PC9 (HB), sensor_pin: sk1:PC4 (THB), fan sk1:PC7 (FAN1).
- **Tools 6–7** on sk2, the same pins.
- [pin_watch io]: add pin_t4: ^!sk1:PC0, pin_t5: ^!sk1:PC1 (and sk2 for 6–7). pin_watch uses Klipper's buttons module, which reads pins on any board.
- [duplicate_pin_override]: add the new dock-switch pins to its pins: list, as Sergei's file does for his.
- MHC_variables.cfg: one variable_x_tN per extra dock, and variable_max_tool.

⚠️ Sergei has tested **4 tools**; more is untested. Bring the tools up one at a time (Chapter 5) and run the stress test before printing with all of them.

✅ **Check:** FIRMWARE_RESTART comes up with no errors. Mainsail shows every heater, thermistor and fan, and one tool button per tool.

Chapter 5: First start and calibration

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Work down this chapter **in order**. Keep a hand on the power switch for every first move.

Step 5.1: Motors, one at a time

Do:

1. Power off, plug the motors back in, power on.
2. STEPPER_BUZZ STEPPER=stepper_x, then stepper_y, stepper_z, stepper_z1, stepper_z2, extruder. Each motor buzzes back and forth.
3. **Z:** move Z +10 mm. All **three** screws must move the bed the **same way**. If one goes the wrong way, flip its dir_pin (add or remove !).
4. **CoreXY:** toolhead in the middle, move **+X**: it must go **right**; move **+Y**: it must go to the **back** (toward the docks). If not, swap the A/B sections or flip a dir_pin as the Klipper docs describe.

✅ **Check:** every motor moves the right way at low speed.

Step 5.2: Homing and travel (with no tool on the toolhead)

Do:

1. Tune sensorless homing (Sergei's sensorless.cfg): home X and Y **5 times**, and the position must be the same every time.
2. Jog slowly to each end: set position_min and position_max for X and Y. **Y max** must reach the docks (Step 5.5) without touching anything.
3. Eddy: measure its X/Y offset; run PROBE_EDDY_CURRENT_CALIBRATE and the tap calibration (BTT's Eddy guide).
4. Z_TILT_ADJUST; repeat until the range is under 0.01 mm. Then BED_MESH_CALIBRATE.

✅ **Check:** homing is repeatable, and no move reaches a hard stop.

Step 5.3: Heaters and fans, one tool at a time

Do (each tool in turn, starting with tool 0):

1. Fit **only this tool** on the toolhead. The others stay in their docks.
2. PID_CALIBRATE HEATER=extruder TARGET=220 (then extruder1, extruder2, ...). SAVE_CONFIG.
3. Its 3010 fan comes on above 50 °C ([heater_fan]).
4. Check its **dock switch**: in Mainsail, its tool button turns green when the tool is in its dock and red when it is out (pin_watch).
5. PID_CALIBRATE HEATER=heater_bed TARGET=60 once.

✅ **Check:** every heater holds its temperature, every fan runs, and every dock switch changes state.

Step 5.4: The feeder servo

Do:

1. **With the linkage disconnected**, move the servo slowly with SET_SERVO SERVO=my_servo ANGLE=... to find the angle for **open** and for **closed**.
2. Connect the linkage at the closed angle. Step slowly to open and back, watching for binding or over-travel.
3. Write both angles into variable_servo_open_angle / variable_servo_close_angle and restart.

✅ **Check:** the idler opens and closes fully, without the servo straining at either end.

Step 5.5: The dock coordinates (rear docks)

Do (each dock):

1. Home. With **no tool** on the toolhead, jog over **dock N** at a safe height.
2. Line the toolhead's guide pins up with the parked tool's receptacles. Read **X** from Mainsail: that is variable_x_tN. They should be about **58 mm** apart.
3. Jog slowly toward the back: find **Y** just clear of the tools (variable_y_safe) and where the pins are fully in (variable_y_latch).
4. Restart Klipper after editing MHC_variables.cfg.
5. Pick up and park each tool with **slow speeds** first (variable_slow_speed, variable_fast_speed lowered), then at Sergei's values.

✅ **Check:** every tool is picked up and parked cleanly, 10 times in a row, at full speed. Watch for a dock switch that doesn't change state.

Step 5.6: Stress test, offsets and the slicer

Do:

1. Run a long series of tool changes (the community's TEST_TOOLS stress-test macro, or your own loop of T0 … TN) before the first real print.
2. Tool offsets: measure X/Y/Z between the tools. **MedusaHC-Calibrate** does this automatically (contact sensor, or Eddy tap for Z), or measure by printing.
3. Slicer: import the OrcaSlicer printer profile from MedusaHC's Slicer folder. Set the bed size and tool count for your frame. Use START_PRINT for the start G-code (MedusaHC issue #2).
4. Input shaping: SHAPER_CALIBRATE with an ADXL345. If you fit the frame upgrades, compare with the upgrades manual's table.
5. Pressure advance, flow and temperature: Ellis' Print Tuning Guide.

✅ **Check (end of manual):** a multi-colour test print (MedusaHC's Slicer 3MF) finishes with clean tool changes. No CHANGE_ME or placeholder values are left in the configuration.